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HIGHER EDUCATION MODULE GUIDE

2025/26

Curriculum Area:	Computing
Programme Name:	BSc (Hons) Computing [Top-up]
Module Code & Name:	CMP600 – Dissertation
Module Tutor Name:	Anjuna Deva Raj
Moderator Name:	Dr Raphael Ejime
Moderation Date:	22/07/2025

Module Overview

Welcome to Dissertation CMP600

This module enables students to undertake a significant and sustained piece of independent work in computing. Students will explore a topic of interest through primary or secondary research, developing a substantial artefact and academic documentation. This journey promotes deeper understanding of academic, technical and professional practice, with scaffolded support through supervision, workshops, and online resources.

Aims and Learning Outcomes of the Module

The focus of this module is learning underpinned by research. Students will develop skills in research and analysis through independent study of a related topic of their choice. Critical thinking skills will be developed along with an ability to design and construct research and then complete a project relating to their topic.

Learning Outcomes

- **LO1:** Independently apply research, analytical, and project management skills to plan, execute, and deliver a substantial independent project

- **LO2:** Analyse and critically evaluate the reliability and significance of current thinking/research in a chosen field

- **LO3:** Apply theory to practice to generate solutions to complex problems both within the project itself and its management

- **LO4:** Demonstrate judgment in being able to choose and justify solutions to the problems posed within the project and in its management

- **LO5:** Critically reflect on personal learning and skill development gained through independent research, demonstrating self-awareness of strengths and areas for improvement.

Module Plan

Semester 1

Week	Topic	Self-Directed Study
1	**Module Launch, Topic Exploration**	Begin to address sources for preliminary research and ideas for your topic area. You should be identifying sources that will help to provide a theoretical and technical background on your chosen approaches. Things to do: Identify a topic using academic databases (IEEE, ProQuest). Begin project idea log. Read: Cottrell (2021), Greetham (2019).
2	**Ethics Workshop, Proposal Development Ethical Approval Process**	Begin to address sources for preliminary research and ideas for your topic area. You should be identifying sources that will help to provide a theoretical and technical background on your chosen approaches. Provide guidance on ethics form. Read: Creswell (2018) on research design. Draft initial objectives. Review VLE sample proposals.
3	**Proposal Workshop**	Dissertation Supervision Meeting 1, Using the techniques shown today and peer-resources, begin to structure your proposal, using the research you have completed so far to assist you. Read: Bell & Waters (2018) on proposal writing.
4	**Proposal Workshop** **Dissertation Supervision Meeting 1**	Begin to address sources for preliminary research and ideas for your topic area. You should be identifying sources that will help to provide a theoretical and technical background on your chosen approaches. IT Guidelines: Set up Trello/Jira. Watch VLE videos on project scoping.
5	**Agile Methods, Gantt Planning**	Using the techniques shown today and peer-resources, begin to structure your proposal, using the research you have completed so far to assist you. Focus on to create project plan (WBS + Gantt). Explore tools like Miro, Notion. Read: Sommerville (2016), Project Management Institute (PMI) guides.
Academic Development Week		Following your supervisor meeting, reflect on any feedback provided and action anything that has been highlighted.
6	**Sprint Planning (Sprint 1)**	Begin to transfer your proposal to the pitch template. Consider how you will convey information to your audience.

		Create Trello board with sprints. Read on agile workflows: [SCRUM model for agile methodology](#) , [Atlassian Agile Guide](#).
7	**Proposal Submission Friday 5th December 2025 by 4pm**	Following your supervisor meeting, reflect on any feedback provided and action anything that has been highlighted. Discuss the ethics form with supervisor for any update Things to do: Present project plan and early findings. Reflect on proposal development, Include the ethics form along with your proposal and submit via Turnitin.
8	**Research Methods Workshop**	Continue literature review. Use [Microsoft Support](#) for Bibliography, citation and referencing.
9	**Sprint 1 Planning** Supervisor Tutorials Supervisor Meeting 2	Following your Progress Review, look over your feedback carefully to identify any issues that need resolving. In particular, if there are any issues highlighted in feasibility, ensure that you properly scope your project idea. Things to do: Submit refined problem statement and methodology.
Xmas Break		During these weeks, you will have scheduled one-to-ones with your supervisor. Please make sure you are prepared and have any required documentation/practical work for review. Your sprints should be properly documented. Ensure that planning and retrospectives are completed correctly and that your sprint boards are addressed appropriately
Xmas Break		
10	**Sprint 1 Workshop** Supervisor Tutorials	Set up your Trello board utilising your preferred strategy (agile, Kanban etc.) Things to do: Begin early implementation/testing or design prototyping.
11	**Technical Design Reviews**	Draft early artefact features. Study: [Gamma et al. (1995) on software design patterns.](#)
12	**Supervisor Meeting 3**	Review artefact progress. Prepare for progress presentation.
13	**Sprint 2 Workshop** Supervisor Tutorials Progress Presentation (Formative)	During these weeks, you will have scheduled one-to-ones with your supervisor. Please make sure you are prepared and have any required documentation/practical work for review. Your sprints should be properly documented. Ensure that planning and retrospectives are completed correctly and that your sprint boards are addressed appropriately Things to do: 1. Deliver project update. Reflect on challenges and feedback. 2. Submit sprint review. Adjust backlog and planning. 3. Begin drafting full documentation: Read: Bailey (2018), Swales & Feak (2012).
14	**Sprint 2 Completion and Retrospectives** Supervisor Tutorials	
15	***Formative – Progress Review Presentations*** **W/C [02/02/2026 - 06/02/2026]**	
16	***Formative – Progress Review Presentations*** **W/C [09/02/2026 - 13/02/2026]**	

Half Term		Following your Progress Review, look over your feedback carefully to identify any issues that need resolving. If required, adjust your sprint boards appropriately to reflect changes

Semester 2

Week	Topic	Self-Directed Study	
1	**Sprint 3 Planning**	Deepen experimentation. Explore advanced techniques.	During these weeks, you will have scheduled one-to-ones with your supervisor. Please make sure you are prepared and have any required documentation/practical work for review. Your sprints should be properly documented. Ensure that planning and retrospectives are completed correctly and that your sprint boards are addressed appropriately
2	**Supervisor Meeting 4**	Share updated artefact and logs. Reflect on setbacks.	
3	**Sprint 3 Execution**	Test, document iterations. Read: IEEE Xplore, Case Studies.	
4	**Research Writing Workshop**	Draft literature review and methodology sections.	
5	**Supervisor Meeting 5**	Check alignment with LO2 and LO3. Iterate documentation.	
	Easter Break		
	Easter Break		
6	**Sprint 4 Completion and Retrospectives** Supervisor Tutorials-**Sprint 4 Planning**	Things to do: Prepare user feedback or evaluation protocol.	During these weeks, you will have scheduled one-to-ones with your supervisor. Please make sure you are prepared and have any required documentation/practical work for review. Your sprints should be properly documented. Ensure that planning and retrospectives are completed correctly and that your sprint boards are addressed appropriately.
7	**Sprint 5 Planning** Supervisor Tutorials-Usability Testing / Artefact Review	Analyse data or outcomes from user feedback.	
8	**Sprint 5 Workshop** Supervisor Tutorials- Supervisor Meeting 6	Final check-in. Plan final edits.	
9	**Sprint 5 Completion and Retrospectives** Supervisor Tutorials-Final Write-Up Support	Submit draft to peer/supervisor. Review feedback.	
10	**Project Workshop**- Editing & Proofreading Workshop	Review grammar, referencing, structure. Use Grammarly or Hemingway Editor.	
	Half term		
11	**Final Project Submission (Documentation + Artefact)**		

	[22/05/2026]	Upload to Turnitin. Zip artefact. Provide Trello link.
12	**Viva Voce**	Times to be organised with your teacher.
13	**Exit Tutorials**	
14	**Exit Tutorials**	Reflect on feedback and lessons learned. Plan progression.

Note: The delivery plan may be subject to minor amends in response to the progress and pace of the student group.

Assessment

Assessment of students' work on this programme adheres to the NCG HE Academic Regulations. A full set of regulations are available by clicking the link below.
[HE Regulatory Documents - NCG (ncgrp.co.uk)]

Formal Formative Assessment
Formal formative is a method of assessment used to support your development and progress you are making within this module. Formal formative assessments are not graded; however you will receive feedback from your module lead to support with the summative assessment (graded assessment).

Assessment Tool	Deadline/Method of Submission
Ongoing Supervisor Tutorials	**Meeting 1 deadline** on W/C **03/11/2025 - 07/11/2025** **Meeting 2 deadline** on W/C **15/12/2025 - 19/12/2025** **Meeting 3 deadline** on W/C **19/01/2026 - 23/01/2026** **Meeting 4 deadline** on W/C **09/03/2026 - 13/03/2026** **Meeting 5 deadline** on W/C **30/03/2026 - 03/04/2026**
Project Proposal	**Deadline Date:** **Friday 5th December 2025 by 4pm** via Turnitin
Project Progress Review	**Deadline Date:** W/C **[02/02/2026 - 06/06/2026]**, W/C **[09/02/2026 - 13/02/2026]**, submitted via Turnitin

Formative Assessment Overview

Ongoing Supervisor Tutorials

Schedule one-to-ones across the academic year to help support your project

Project Proposal

PowerPoint presentation (10 minutes max.) All students must create a short PowerPoint presentation that details their project proposal. An outline of the content and expectations will be provided in class. Students will present their presentations to the group for feedback from peers and lecturer.

Project Progress Review

As a part of the normal one-to-one tutorial process, you will present your work-in-progress Project Documentation draft.

Summative Assessment

Summative assessments are mandatory and administered to assess your knowledge and understanding of the learning outcomes of the module. Completion of summative assessments will result in a grade which contributes towards your overall grade of your programme.

Assessment Tool	% Weighting	Learning Outcomes Covered	Deadline/Method of Submission	Feedback Date (20 working days post submission)	Re-submission Deadline/Method of Submission
Project	**100%**	**L0's 1, 2, 3, 4, & 5**	**Friday 22nd May 2026 by 4pm** via Turnitin	**Friday 26th June 2026**	**Friday 24th July 2026 by 4pm** via Turnitin

Assessment Overview

The assessment for this module is Project 100%. This is split into three (3) elements with their weightings. The final module grade will be the weighted average of each of the respective grades. Please see *Assessment Elements* below for details on each element.

Extended Project

An Extended Project involves the development of a substantial artefact, informed by broad research of current industry standards and practices. Your project will be supported by project documentation that demonstrates contextual research, planning and design, experimental work, development and project management.

Your project should focus on a specific problem relating to an area of interest within Computing, including Networking, Cyber Security or Software Engineering. A proposal must be completed to outline your intentions and presented for approval.

For all project types, you are to present a Proposal to the group (see Formal Formative Assessment above).

Supervisor Tutorials

The module is designed for you to apply the skills you have developed within your previous study along with self-initiated development of new, relevant skills, to form a novel product. Whilst there is no delivered content for this module, you will be entitled to a total of 6 45-minute tutorials with your supervisor for guidance and facilitation of your project. Notes will be stored in the Class Notebook. Depending on class size and timing, your meetings may be broken into more, shorter tutorials across the year.

Assessment Elements

<u>Project Documentation (40% Weighting)</u>

For all project types, appropriate documentation and project management should be produced to demonstrate your undertaking of the project. You must maintain project documentation and collate it into a single document for submission (Word/PDF).

Contextual Research

Your project documentation should incorporate a research aspect. For your project type, you must demonstrate a clear and broad understanding of the approaches, software, and techniques you could employ within your work. This should be presented in your documentation with a clear reference to contemporary industry practitioners, software documentation, or academic literature. All work should be appropriately cited and referenced. There should also be an experimental focus, demonstrating the application of theory to your practice. It should be evident how what you have looked at may have informed your project.

Project Planning and Design

Your project documentation (approx. 7,000 words), should include all design work. The context of your project will dictate the type and nature of your software/product, but you will find a non-exhaustive list below for some suggestions:

- Project Title, background and motivation, project objectives, and scope
- Requirements analysis (functional and non-functional, stakeholder analysis, if any)
- Project Planning (WBS, Gantt Chart, Resource Planning, and Risk Assessment)
- Methodology (System Design Diagrams, Tools and Technologies, Data Requirements, and Implementation Steps)
- Security Considerations
- Results and Evaluation (Performance Metrics, Testing Strategy, Challenges)
- Project Management and Version Control
- References

Your work should present a highly considered approach to the project, with clear evidence of iteration and development based on feedback and exploration of your ideas.

You must maintain appropriate planning for your project. Projects should be run in an agile fashion, with Trello/Jira used to plan and organise your project. Each sprint should have a planning card that outlines planned activities, each task card taken from the backlog, and a retrospective card at the end of the sprint to evaluate your progress. Add your supervisor to your Trello/Jira board and ensure a link is provided in your documentation.

Project Product (40% Weighting)

You are tasked with devising and developing your final proposed Product. You may use any software/tools you wish to complete the work.

Your final Project Product must be suitably presented in the form of a working prototype or system that addresses a real-world problem in software, networking, or cybersecurity. For software solutions, you should have an executable built and uploaded to repository, for example, GitHub.

Viva Voce (20% Weighting)

Upon completion of your Project, you are required to attend a Viva Voce, in which you will present, discuss, and defend areas of your Project. The Presentation will involve a series of questions that further assess your engagement and management of the Project. A structured brief will be provided, with Presentations expected to last 30-45 minutes.

SUGGESTION - Submit your Project Documentation as a Word/PDF document to the Documentation Turnitin portal.

Submit your Project Product within a ZIP folder to the Product Turnitin portal

Viva Voce will be scheduled with timeslots posted to the VLE.

Learning Outcomes Covered - 1, 2, 3, 4, 5

Academic Submission Expectations

Reports/ Portfolios / Projects

- 1.5 line spacing
- Font: Arial
- Font size: 12
- Sub-headings presented in bold and/or underlined
- Page numbers required
- Text presented to a left alignment
- Reference list and bibliography presented in alphabetical order
- Front cover to include name, student name, module code and title, module leader and submission date.

Presentations

- Appropriate software suitable for a presentation
- Font size, font type and colour has flexibility
- Reference list/bibliography alphabetically presented
- Will be required on a memory stick for presenting
- Appropriate colour scheme
- Formal clothing should be worn to present

VIVA/Interview Assessment

- Verbal assessment with no notes present
- Individual time slots will be presented
- No questions can be asked to an assessor during assessment
- Formal clothing should be worn to attend

Contextualised Grading Criteria

Grading Bracket	85-100%	70-84%	60-69%	50-59%	40-49%	30-39%	0-29%
Project Documentation (40% Weighting)	Exceptional depth and breadth of research, demonstrating originality, critical insight, and an innovative application of theory to practice with clear impact on project outcomes. Meticulously crafted, industry-level documentation that is exemplary in clarity, organisation, planning detail and design execution, demonstrating leadership in project development.	Comprehensive, well-integrated research from academic, practitioner, and technical sources, critically analysed and effectively applied to practical development. Highly coherent and professional documentation, with excellent planning, risk assessment, iterative design processes, and clear technical specifications.	Strong research base including practitioner and technical sources, with good application of theoretical concepts to project development and reflective experimentation. Detailed and logical planning and design documentation, showing clear understanding of processes, iterative design, and	Research demonstrates awareness of relevant theory, software, and techniques, with some evidence of practical experimentation and synthesis. Project planning and design documentation is functional, with appropriate structure and moderately justified decisions. May	Some relevant research is presented but lacks depth or critical engagement. Application of theory to practice is attempted but underdeveloped. Basic project planning and design documentation is included but may be overly simplistic, lacking clear rationale or forward-thinking.	Basic or inaccurate attempts at research, with weak understanding and very limited engagement with sources or concepts. Lacks application of relevant theory or process. Disorganised or incomplete documentation, with unclear or impractical project planning and	Minimal or no evidence of relevant research, with poor or absent understanding of theoretical and practical concepts. Practitioner or software research is missing or irrelevant. Little to no project planning or design documentation; what is provided lacks coherence, structure, and

			alignment with project goals	lack refinement.		minimal design thinking.	alignment with any stated goals.
Project Product (40% Weighting)	Pioneering solution with unique technical/creative elements. Exceeds all requirements with flawless execution and advanced features. Professional, intuitive UI with exceptional accessibility/UX. Efficient, modular code/configurations with full documentation/comments.	Creative approach; stands out from standard solutions. All core functions work perfectly, minor enhancements possible. Clean design; minor usability improvements needed. Well-structured code/configs with minor documentation gaps.	Some novel ideas but largely conventional. Core functions operational but may contain minor bugs. Clean design but with noticeable navigation issues. Basic structure with inconsistent commenting.	Minimal creativity; follows existing/familiar templates. Basic functionality achieved; significant bugs or missing features. Confusing layout; usability barriers exist. Messy code with minimal documentation.	Generic solution lacking originality. Limited functionality; only partial core features work. Poor design; difficult to use. Hard-to-maintain and lacking organisation.	No innovation attempted. Barely functional; minimal requirements met. Minimal effort; largely unusable. Sloppy or incomplete code.	Non-functional or not submitted. No design consideration. No code submitted.
Viva Voce (40% Weighting)	Exceptional presentation quality, combining clarity, confidence, polish, and depth. Demonstrates	Engaging, professional, and articulate presentation with excellent	Well-delivered, confident presentation that communicates	Clear and structured presentation that outlines the product	Basic explanation of the project and process, covering	Disorganised or superficial presentation, with limited explanation of	Incoherent or incomplete presentation, lacking structure,

	outstanding communication and storytelling. Sophisticated, original critical insight, with a high level of reflexivity, synthesis of research and practice, and clear articulation of lessons learned and broader implications.	clarity, structure, and timing. Tailored well to the audience. Insightful, evidence-based critical reflection that connects practice with research and shows deep awareness of the development journey, challenges, and creative/technical decisions.	project goals, outcomes, and process effectively to an informed audience. Thoughtful and well-supported critical analysis, showing awareness of successes, limitations, and the influence of research or feedback on decisions.	and development process with reasonable clarity and logical flow. Moderate level of reflection, identifying key decisions and challenges with some justification; beginning to link theory and practice.	essential elements but with shallow or uneven depth. Some attempt at critical reflection, but insights may be vague, unsupported, or unconvincing; weak articulation of learning or outcomes.	the project or rationale behind decisions. Minimal critical analysis, showing poor understanding of methods used or challenges faced; heavy reliance on description rather than reflection.	clarity, or relevance to the project. Shows minimal preparation or engagement. No critical reflection or insight into development decisions, challenges, or outcomes; no clear understanding of the product or process.

Grading Bracket	Project Documentation (40% Weighting)	Project Product (40% Weighting)	Viva Voce (40% Weighting)				
85-100%	Exceptional depth and breadth of research, demonstrating originality, critical insight, and an innovative application of theory to practice with	Pioneering solution with unique technical/creative elements. Exceeds all requirements with flawless execution and advanced features. Professional, intuitive UI with exceptional accessibility/UX.	Exceptional presentation quality, combining clarity, confidence, polish, and depth. Demonstrates outstanding communication and storytelling.				

	clear impact on project outcomes. Meticulously crafted, industry-level documentation that is exemplary in clarity, organisation, planning detail and design execution, demonstrating leadership in project development.	Efficient, modular code/configurations with full documentation/comments.	Sophisticated, original critical insight, with a high level of reflexivity, synthesis of research and practice, and clear articulation of lessons learned and broader implications.				
70-84%	Comprehensive, well-integrated research from academic, practitioner, and technical sources, critically analysed and effectively applied to practical development. Highly coherent and professional documentation, with excellent planning, risk assessment, iterative design processes, and clear technical specifications.	Creative approach; stands out from standard solutions. All core functions work perfectly, minor enhancements possible. Clean design; minor usability improvements needed. Well-structured code/configs with minor documentation gaps.	Engaging, professional, and articulate presentation with excellent clarity, structure, and timing. Tailored well to the audience. Insightful, evidence-based critical reflection that connects practice with research and shows deep awareness of the development journey, challenges, and creative/technical decisions.				
60-69%	Strong research base including practitioner and technical sources, with good application of theoretical concepts to project development and reflective experimentation.	Some novel ideas but largely conventional. Core functions operational but may contain minor bugs. Clean design but with noticeable navigation issues.	Well-delivered, confident presentation that communicates project goals, outcomes, and process effectively to an informed audience.				

	Detailed and logical planning and design documentation, showing clear understanding of processes, iterative design, and alignment with project goals	Basic structure with inconsistent commenting.	Thoughtful and well-supported critical analysis, showing awareness of successes, limitations, and the influence of research or feedback on decisions.				
50-59%	Research demonstrates awareness of relevant theory, software, and techniques, with some evidence of practical experimentation and synthesis. Project planning and design documentation is functional, with appropriate structure and moderately justified decisions. May lack refinement.	Minimal creativity; follows existing/familiar templates. Basic functionality achieved; significant bugs or missing features. Confusing layout; usability barriers exist. Messy code with minimal documentation.	Clear and structured presentation that outlines the product and development process with reasonable clarity and logical flow. Moderate level of reflection, identifying key decisions and challenges with some justification; beginning to link theory and practice.				
40-49%	Some relevant research is presented, but lacks depth or critical engagement. Application of theory to practice is attempted but underdeveloped. Basic project planning and design documentation is included but may be overly simplistic, lacking clear rationale or forward-thinking.	Generic solution lacking originality. Limited functionality; only partial core features work. Poor design; difficult to use. Hard-to-maintain and lacking organisation.	Basic explanation of the project and process, covering essential elements but with shallow or uneven depth. Some attempt at critical reflection, but insights may be vague, unsupported, or unconvincing; weak articulation of learning or outcomes.				

30-39%	Basic or inaccurate attempts at research, with weak understanding and very limited engagement with sources or concepts. Lacks application of relevant theory or process. Disorganised or incomplete documentation, with unclear or impractical project planning and minimal design thinking.	No innovation attempted. Barely functional; minimal requirements met. Minimal effort; largely unusable. Sloppy or incomplete code.	Disorganised or superficial presentation, with limited explanation of the project or rationale behind decisions. Minimal critical analysis, showing poor understanding of methods used or challenges faced; heavy reliance on description rather than reflection.				
0-29%	Minimal or no evidence of relevant research, with poor or absent understanding of theoretical and practical concepts. Practitioner or software research is missing or irrelevant. Little to no project planning or design documentation; what is provided lacks coherence, structure, and alignment with any stated goals.	Non-functional or not submitted. No design consideration. No code submitted.	Incoherent or incomplete presentation, lacking structure, clarity, or relevance to the project. Shows minimal preparation or engagement. No critical reflection or insight into development decisions, challenges, or outcomes; no clear understanding of the product or process.				